

EYAD JAWAD

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EDUCATION

University of Technology (UOT)

B.Sc. in Computer Science (Year 2)

Baghdad, Iraq

Expected 2029

- Class Valedictorian — ranked 1st in cohort during the first academic year.
- Certification: Harvard University's CS50 (Introduction to Computer Science).

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, HTML, CSS, SQL, Yaml

Core Concepts: Data Structures, Algorithms, Bit Manipulation, Memory Management, Image Processing, Asynchronous Programming, 3D Rendering, Backtracking, Authentication & Security, Software Testing, Documentation

Tools & Frameworks: Git, Docker, FastAPI, SQLAlchemy, PostgreSQL, Redis, SQLite, WebGL, Pygame, BeautifulSoup, GTest, PyTest, Node.js/npm, Makefiles

KEY PROJECTS

Telegram Archiver (Python, Telethon) [GitHub](#)

- Built asynchronous streaming data pipelines using Telethon to index and archive 1,000,000+ chat messages in less than 4 hours.
- Implemented a fault-tolerant checkpoint system for precise execution-state persistence and graceful recovery from connection errors.
- Designed relational and SQL schemas ensuring data integrity across multi-format parsing.
- Wrote a comprehensive test suite covering almost all possible cases, with CI/CD matrix testing across Python 3.11–3.14.

Todo Application (Python, FastAPI, PostgreSQL, Docker) [GitHub](#)

- Architected a full-stack task-management service — layered FastAPI backend, vanilla JS frontend, PostgreSQL, and Redis — evolving it from an async terminal client into a deployed multi-container system.
- Implemented user authentication with Argon2 password hashing and rate-limited endpoints, separating the codebase into distinct service, query, and API layers.
- Containerized the backend with Docker and an nginx reverse proxy, deploying the frontend independently via GitHub Pages; reached 94% test coverage across unit and integration suites.

Maze Game (C++, JavaScript, WebGL) [GitHub](#)

- Engineered optimized C++ maze generation and solving algorithms from scratch — generation under 1ms (down from a 517ms baseline) and backtracking-based solving under 15ms (down from 5626ms), a 500× improvement.
- Built a zero-dependency, interactive 3D rendering environment using raw WebGL and native JavaScript.
- Deployed a production web build via GitHub Pages for public, real-time browser-based testing.

Open Source Contribution — grace (Python, pytest) [View PR](#)

- Shipped the `translate_from` feature for grace's `translator_cog` (Code Society Lab's open-source Discord bot), and wrote a shared pytest fixture module (`confest.py`) and raised test coverage from 27% to 29%.
- Added nickname moderation that auto-renames users with inappropriate names to a randomly generated two-word name, with a matching test suite.

Custom Memory Allocator — elloc (C) [GitHub](#)

- Implemented a from-scratch malloc/free replacement backed by an AVL-tree free-list for $O(\log n)$ block lookup in place of a linear scan.
- Engineered chunk splitting and recursive coalescing on free to limit fragmentation, with alignment guarantees and integer-overflow-safe size calculations.
- Extended the process heap on demand via direct `sbrk` syscalls; validated correctness with an accompanying unit test suite.

Image Processing Engine & Technical Research (C++) [GitHub](#)

- Programmed a low-level computer vision pipeline executing Sobel filtering, Gaussian blur, and full Canny edge detection from scratch, including non-maximum suppression and hysteresis tracking.
- Built a custom 2D convolution matrix layer supporting dynamic kernel generation and direct binary parsing of raw BMP files.
- Co-authored a technical research paper evaluating low-level kernel convolutions across 10+ academic sources.

Huffman Compression Tool (C++) [GitHub](#)

- Built an end-to-end compression utility implementing custom Huffman encoding and decoding from the ground up.
- Designed a proprietary binary archive format (.HUF) with strict bit-level packing to eliminate structural overhead.
- Optimized recursive binary tree traversal for deterministic serialized tree packing and unpacking.

ACHIEVEMENTS & AWARDS

- Winner — Excellence Hackathon 2026, University of Baghdad: secured 1st place, out-competing a highly selective group of top-ranked first-year candidates.
- 3rd Place — National Data Science Hackathon 2026, University of Al-Farabi: co-led a 3-person team through a 24-hour national hackathon, placing 3rd of 30+ teams that included seniors, advanced students, and graduates.